## Minimum Transmit Sampling for Resolution Preservation in Ultrasound Tomography

### Background, Motivation, and Objective

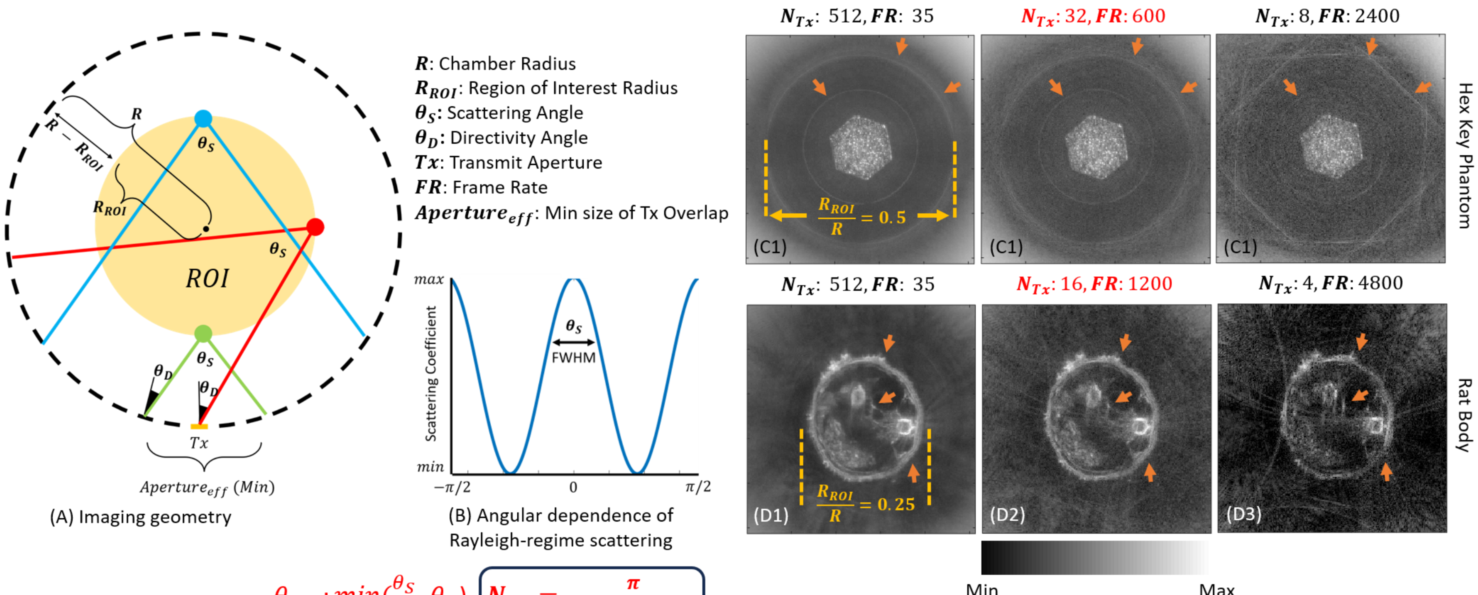
*Reflection ultrasound computed tomography (RUCT) enables high-resolution full-view imaging for breast and small-animal imaging. RUCT typically uses many transmit events to provide sufficient angular diversity for reconstruction. However, dense transmit sampling limits the achievable frame rate (FR), which is critical for ultrafast imaging. Moreover, the minimum transmit sampling required to preserve spatial resolution is not well established, and more transmissions than required may be used. The objective of this work is to determine the minimum number of transmit events required to avoid resolution degradation within a region of interest (ROI), enabling transmit sparsification and higher FR USCT imaging.*

### Statement of Contribution/Methods

*We derive a physics-based criterion to estimate the minimum transmit events required to preserve resolution in ring-array USCT. A circular array surrounds a ROI with radius* $R_{ROI}$*. Due to the combined effects of Rayleigh-regime scattering (Fig. B) and transducer directivity* ($\theta_D$)*, the detectable scattered field from a given transmit element is confined to a limited angular sector characterized by an effective scattering width* $\theta_{eff}$ *(Fig. A), defined as the minimum between the FWHM of the scattering angular response* $\theta_s$ *and the transducer directivity* $\theta_D$*. The effective angular coverage per transmit is further limited by the angular span of the ROI. The effective angular step and resulting minimum number of required transmit events are illustrated in Fig. A,B (red equation).*

### Results/Discussion

*Two datasets were evaluated using a 512-element ring array (R=4 cm, f=5 MHz) with 256 receivers per transmit. For the hexKey phantom with* $R_{ROI}/R=0.5$ *(Fig. C1–C3), the model predicts* $N_{Tx} \approx 32$ *which preserves the structures indicated by arrows. Reducing* $N_{Tx}$ *leads to resolution loss, particularly away from the center. For the in-vivo rat with* $R_{ROI}/R=0.25$ *(Fig. D1–D3), the predicted requirement decreases to* $N_{Tx} \approx 16$*. Reconstructions confirm that features degrade as* $N_{Tx}$ *decreases and as structures lie farther from the center. The agreement between the predicted transmit requirements and the observed resolution degradation across both datasets supports the validity of the proposed physics-based criterion for determining the minimum transmit sampling needed to preserve spatial resolution in sparse-TX RUCT.*




(A) Imaging geometry

(B) Angular dependence of Rayleigh-regime scattering

$$\theta_{eff}: min(\frac{\theta_S}{2}, \theta_D), \boxed{N_{Tx} = \frac{\pi}{\theta_{eff}(1-\frac{R_{ROI}}{R})}}$$

Minimum transmit requirement